

A Computational Framework for Isotopic Analysis in a Nuclear Post-Detonation Environment

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Roman Martinez

University of New Mexico

Department of Nuclear Engineering



GLOBAL &
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POLICY INSTITUTE

Title: A Computational Framework for Isotopic Analysis in a Nuclear Post-Detonation
Environment

Martinez R. Cassiano O. Lamont S.

University of New Mexico, Global National Security Policy Institute

Author Note

0009-0005-0851-2339, <https://orcid.org/0009-0005-0851-2339>

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Rmartinez658@unm.edu

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Abstract

This paper presents an isotopic analysis methodology for spent nuclear fuel using modeled decay curves and empirical reactor data. By simulating neutron flux conditions representative of graphite, heavy water-type reactors, isotopic inventories are computed and compared to measurements from the SFCOMPO 2.0 database. The approach enables interpretation of decay signatures over time, particularly for fission products and actinides, offering insights into irradiation history and burnup. Modeled inventories are generated using PyNE with neutron flux derived from reactor burnup data. This framework supports non-destructive fuel characterization and nuclear forensics through isotope ratio diagnostics.

Keywords: Nuclear Forensics, decay, MCNP, PyNe, Post-Detonation, Ratio

1. Introduction

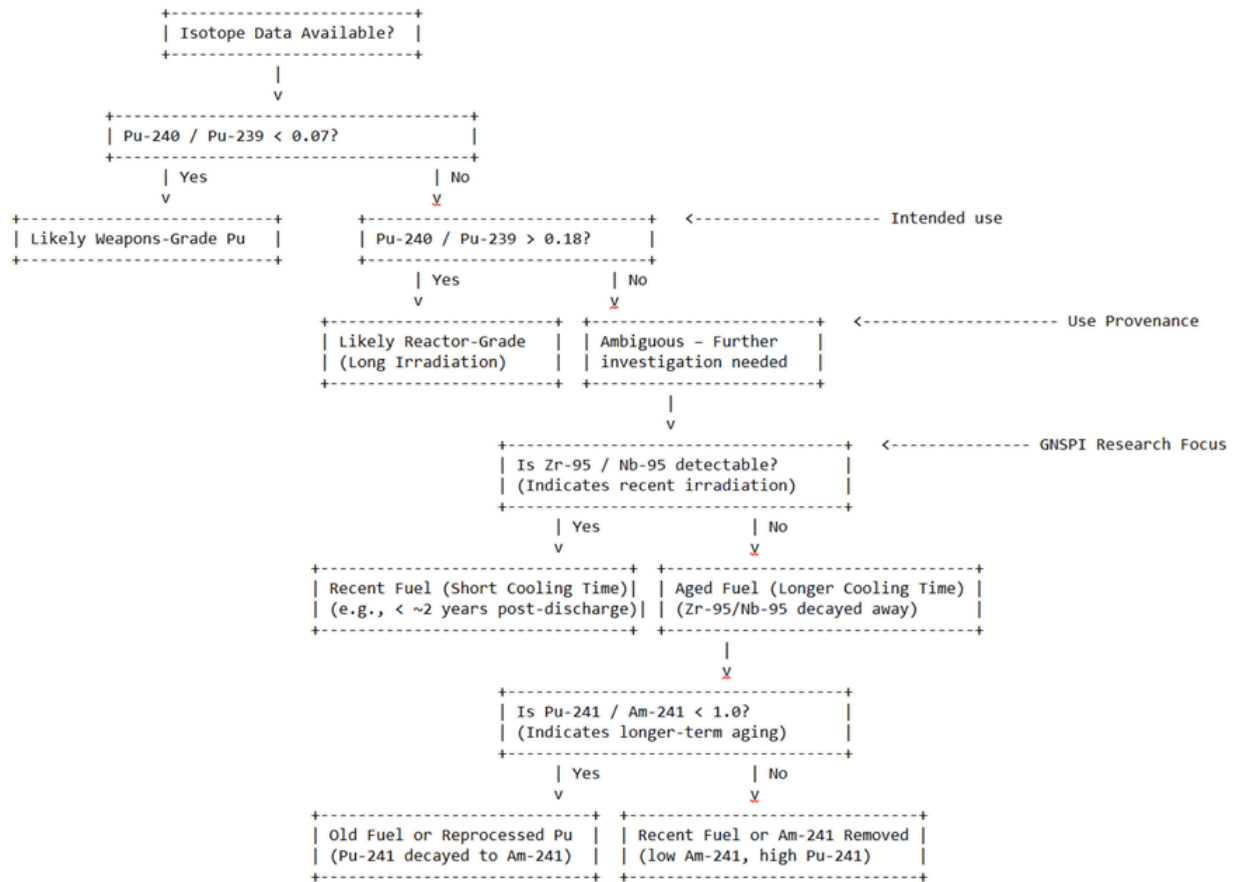
1.1. Purpose of Study

This study was motivated by the noticeable absence of nuclear forensics content in the undergraduate Nuclear Engineering curriculum at the University of New Mexico. The project aims to explore the use of PyNe as a tool for nuclear forensic analysis by simulating post-irradiation isotopic ratios, which can be used to estimate fuel age. These ratios serve as critical indicators of both the time since irradiation and, indirectly, the reactor type and origin of the nuclear material. Since reactor fuel undergoes closely monitored irradiation cycles, the observed isotopic composition can provide valuable clues in identifying and attributing nuclear materials in a forensic context [6-7].

1.2. Background

Here within this study the scope of reactor fuel analyzed from the SFCOMPO 2.0 database was limited to graphite and heavy water moderated reactors. The availability of data made these reactors preferred [8]. In selection of these reactors the following criteria had to be met; Known burnup, discharge date, and fuel measurement date.

While there are many isotopic ratios that lead credence to a model, a few select isotopes were selected specifically for this study. Plutonium-241, Americium-241 and Zirconium-95 were selected. These isotopes were chosen for the express reason that their presence in a nuclear detonation nuclear environment is a indicator of origin, material and most importantly provides a timeline of how old this material could be. All indicators of use provenance which would serve decision makers to respond to events accordingly [3-5].

Figure 1: Isotopic Decision Tree & GNSPI Work Focus.

Nuclear Forensics is the study of nuclear material to determine their origin and history, often in the context of law enforcement or national security applications. The importance of the Plutonium ratio early on distinguishes whether the measured isotopic ratio at an event site is a state actor or from purloined fuel rods [6-7]. Since around 20% of Plutonium-240 is synthesized in fuel rods from commercial reactors there is difficulties faced by nefarious actors that would plan to steal spent fuel rods to create a workable plutonium bomb [9].

Further down the decision tree, the focus on ratios that can further distinguish origin using the PyNe toolkit. The ratio of Zirconium and Niobium 95 is simulated first, with the Plutonium and Americium 240 decay chain simulated and compared with empirical values from the spent fuel database SFCOMPO 2.0.

2. Methodology

2.1. MCNP Simulation Setup

A simplified model of a fuel pin for the RBMK, and MAGNOX are simulated using the Monte Carlo N-Particle transport code (MCNP) with the goal of obtaining a neutron flux output [2].

The MCNP model takes the geometry and fuel composition data from SFCOMPO 2.0 for the construction of the input file and are then fed into PyNe and its Transmutation solver. Table 3 details the neutron flux output from the models (Figures 3-4).

Figure 3: *MCNP MAGNOX Fuel Pin Model.*

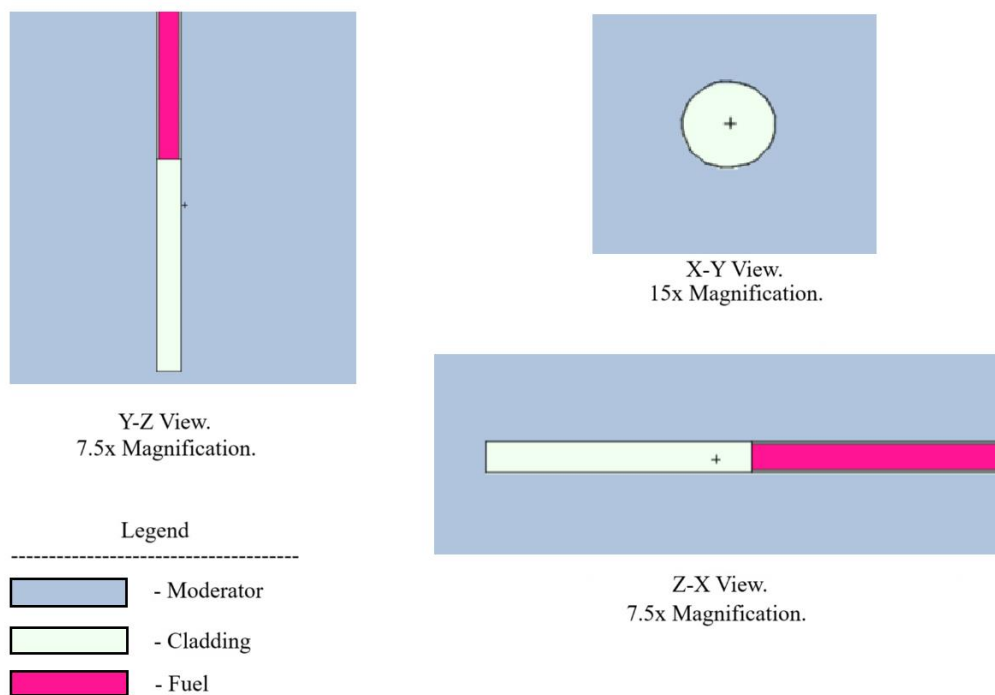
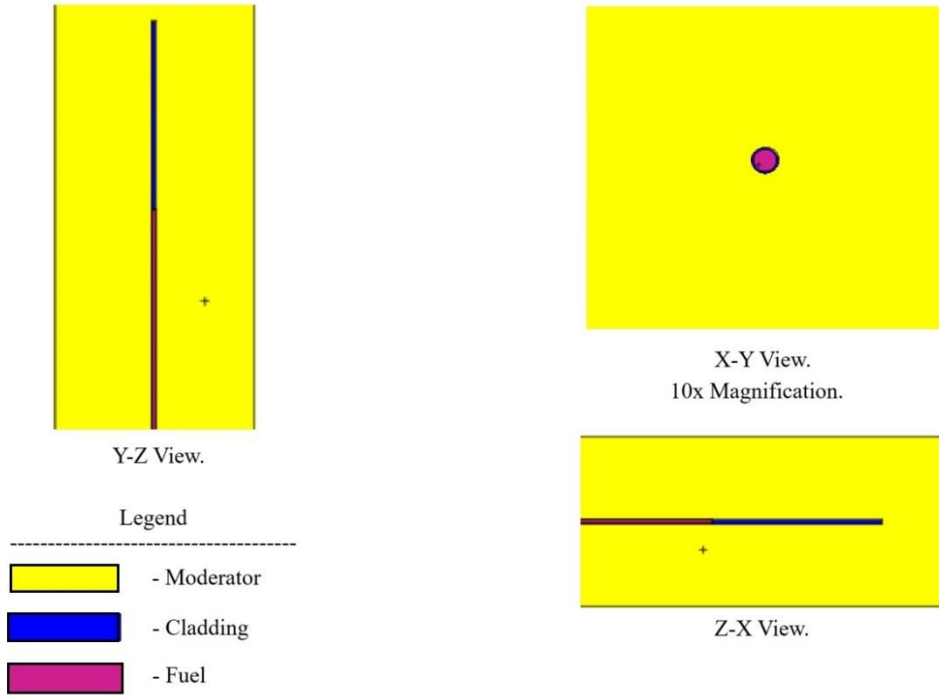


Figure 4: MCNP RBMK-1000 Fuel Pin Model.



2.2. PyNe Workflow

2.2.1. Initial Inventory Calculations

PyNe's ability to simulate how nuclear materials evolve by using the canonical Bateman equation that are internal to PyNe's solver. Equation 1, 1a, 1b and 2 are those that are used in PyNe's solver [1]. Table 1 details the symbols and their respective definitions.

$$N_c(t) = \frac{N_1(0)}{\lambda_c} * \gamma * \sum_{i=1}^c \lambda_i c_i e^{-\lambda_i t} \quad (1)$$

Where the term c_i is defined in Equation 1a:

$$c_i = \prod_{j=1, i \neq j}^c \frac{\lambda_j}{\lambda_j - \lambda_i} \quad (1a)$$

Lastly where γ is defined as the product of the branch ratio between any two species in Equation 1b:

$$\gamma = \prod_{i=i}^{C-1} \gamma_{i \rightarrow i+1} \quad (1b)$$

Table 1. Bateman Equation Symbol Definitions

Symbol	Definition
C	Length of decay chain
i	Index for ith species, on range $[1,C]$
t	Time [seconds]
j	Index for jth species, on range $[1,C]$
$N_i(t)$	Number density of the ith species at time t
λ_i	Decay constant of ith species
γ	Total branch ratio for this chain

Where Figure 2 is our Zirconium-Niobium-95 decay using PyNe. A decay of Zirconium-95 is presented to demonstrate PyNe graphically and the production of daughter products in an idealized decay case. It should be noted that the Zr-Nb ratio uses the pure decay model in PyNe and neglects transmutation caused by neutron flux in a reactor environment.

Figure 2: Decay chain of Zirconium-95 in the instant fission case. Idealized. 200 days.

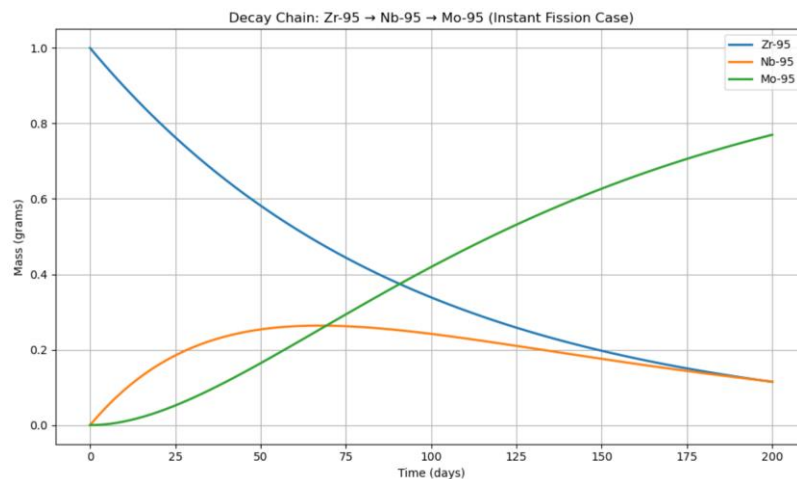


Table 2. *Zr/Nb Ratio. SFCOMPO & Simulated Values.*

Identifier	SFCOMPO Ratio	Simulated Ratio
BRD-1 1 4 D D MAGNOX	6.90	0.88
HUA-1 1 A A MAGNOX	2.09	0.59

2.2.2. Increased Fidelity Calculations

Using simplified models of the respective reactors fuel pins and or rods, we can extract an neutron flux output to then plug into PyNe and obtain an *increased fidelity isotopic ratio*.

Table 3 details the neutron flux output from these models and the isotopic ratios for Plutonium-241 and Americium-241 PyNe models for the RBMK-100 and MAGNOX Reactors [2] [8]. The MCNP script that was written to construct a fuel rod for the respective reactor can be found in the Appendices Section.

Table 3. *Neutron Flux Outputs*

Reactor Type	Neutron Flux (n/cm ² *s)	Model Identifier
RBMK-1000	1.67E13	LEN-1 4-20-3675-85 13 35
MAGNOX	4.4E13	BRD-1 1 4 D D

3. Results

Results from use of PyNe and MCNP produced results that were near agreement with empirical values from the SFCOMPO database with a noticed trend for the lower burnup reactors (<9GW/dUi) that their isotopic ratios aligned with empirical measurements taken. MCNP calculations for the increased fidelity model also produced results that were aligned with the empirical database as these models included the neutron flux replicating a simplified

environment that these isotopes would be present in. Table 4 presents the simulated isotopic ratios with the respective code for the simulations found in the Appendix section of this report.

Table 4. *SFCOMPO 2.0 & PyNE Comparison Results.*

Sample/Bundle Identifier	SFCOMPO Ratio	PyNe Ratio	MCNP + PyNe Ratio
Magnox A	8.04± 0.21	6.65	9.46
Magnox B	9.86 ± 0.34	7.79	6.67
CANDU (PIK-1)	0.92 ± 0.20	0.46	4.6
RBMK1000 (4-20-3675-85 Sample 35)	13466.67 ±188.54	4.91	9.46

3.1. Discussion

3.1.1. Isotopic Buildup and Decay

Neutron flux from the MCNP models is used in complement with PyNe's transmutation equations that take the form of Equation 1 in section 2.2. This neutron flux serves as a direct input into PyNe's transmute function, which allows the use of prediction of isotopic buildup and decay for key fission products and actinides, which is then compared to empirical data from SFCOMPO.

3.1.2. Discussion of Results

The PyNE-MCNP computational framework, as presented in this study models the comparison between our modeled decay curves and the empirical data from the SFCOMPO 2.0 database, which demonstrates a strong qualitative agreement. However, quantitative discrepancies were observed, which can be attributed to several methodological simplifications implemented for rapid iteration.

Firstly, the MCNP fuel pin model employed mirror boundary conditions, which, investigated the neutron flux for a small part of the respective reactor core. While this approach is computationally efficient, it does not fully represent the complexities of an

infinite lattice or the neutron thermalization effects in a complete core. Further investigation to determine how conservative the mirror and periodic boundary conditions are and a more detailed lattice model would provide a more accurate neutron flux, thereby refining the transmutation rates in the PyNE solver. Secondly, the PyNE workflow, while robust, relies on the CINDER cross-section data library. Although compatible, newer data libraries such as ENDF/B-VIII.0 are available and could offer more up-to-date and accurate nuclear data, potentially reducing the observed discrepancies. Finally, the analysis was constrained by the available SFCOMPO 2.0 data, which, in some cases, lacked a full suite of measurement dates or uncertainty values. If a data point was not reported with uncertainties, the uncertainty from the daughter (or parent) was taken as the measurement uncertainty. The propagation of these empirical uncertainties is a critical component of model validation, and its absence for certain data points limits a full statistical comparison.

These factors highlight that while the current framework is effective for demonstrating proof-of-concept, significant opportunities exist for increasing its fidelity.

4. Conclusion

It is shown that with reasonably high confidence that reactor isotopics can be simulated with open-source toolkits with simplified models for a nuclear forensics application. While limitations were encountered during the investigation pertaining to the models, databases and simplifications, future work would focus on the improvement and refinement of these points.

5. References

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6. Appendices

6.1. Appendix A: MCNP Code

```

GNSPI Fuel Rod
c Simplified MCNP input file for fuel rods. Edit as needed.
c =====
c CELL CARDS
c =====
10 1 -10.40 -1 4 -5      imp:n=1 $ UO2 fuel: r<=0.575, z=[0,340]
20 2 -6.45 1 -2 4 -5    imp:n=1 $ Zircaloy sidewall: 0.575<r<=0.68, z=[0,340]
30 2 -6.45 -2 -4 3      imp:n=1 $ Zircaloy bottom endcap: r<=0.68, z=[-12,0]
40 2 -6.45 -2 5 -6      imp:n=1 $ Zircaloy top endcap: r<=0.68, z=[340,400]
50 3 -1.70 -7 2         imp:n=1 $ Graphite: inside RPP, outside outer clad
60 0 7                  imp:n=0 $ Outside problem

c =====
c SURFACE CARDS
c =====
1 RCC 0 0 0 0 0 340 0.575 $ fuel cylinder z=0..340, r=0.575
2 RCC 0 0 -12 0 0 412 0.68 $ outer clad cylinder z=-12..400, r=0.68
3 PZ -12
4 PZ 0
5 PZ 340
6 PZ 400
*7 RPP -30 30 -30 30 -30 552.4 $ 5 ft graphite all around

c =====
c MATERIAL CARDS
c =====
c UO2 (weights)
m1
    92235.70c 0.0209
    92238.70c 0.9750
    92236.70c 0.0038
    8016.70c 0.0003

c
c Zircaloy-2 (weights)
m2
    40090.70c 0.5145
    40091.70c 0.1122
    40092.70c 0.1715
    40094.70c 0.1738
    40096.70c 0.0280

c
c Graphite (trace B-11)
m3
    6000.70c 0.999999
    5011.70c 0.000001

c
c =====
c PHYSICS / SOURCE / RUN
c =====
kcode 50000 1.0 95 235
KSRC 0.0 0.0 1

c
c =====
c TALLIES
c =====
f4:n 10 $ flux in fuel
sd4 1.0

c MCNP code modified for CANDU & MAGNOX fuel rods.
c To change for different reactor config, "mat" and cladding plugs must be changed.

```


6.2. Appendix B: PyNe Code

```

import numpy as np
import matplotlib.pyplot as plt
from pyne import nucname
from pyne.transmute.chainsolve import Transmuter
import uncertainties from uncertainty
# DECAY SIMULATION
# WORKFLOW-----
# Subsequent PyNe scripts follow similar format. Intakes SFCOMPO data.
# Decays material composition and compared to simulated material decay.
# -----

#Calculates the Initial Inventory from Discharge Date
# ---
# Constants and Half-Lives (in seconds)
# ---
half_life_pu241 = 14.3 * 365.25 * 24 * 3600
half_life_am241 = 432.2 * 365.25 * 24 * 3600

lambda_pu241 = np.log(2) / half_life_pu241
lambda_am241 = np.log(2) / half_life_am241

# ---
# MAGNOX Sample A' Data
# ---
# Cooling time from the provided table
cooling_time_years = 0.53
cooling_time_seconds = cooling_time_years * 365.25 * 24 * 3600

# Actual measured values
pu241_measured = 7.117e-5 # g/gU
am241_measured = 8.85e-6 # g/gU

# ---
# Reverse-decay to get initial inventory at discharge
# ---
pu241_initial = pu241_measured * np.exp(lambda_pu241 * cooling_time_seconds)
am241_initial = am241_measured * np.exp(lambda_am241 * cooling_time_seconds)

initial_inventory = {
    nucname.id("Pu241"): pu241_initial,
    nucname.id("Am241"): am241_initial
}

# ---
# Simulate decay to the exact cooling time
# ---
tm = Transmuter(phi=0.0)
result = tm.transmute(initial_inventory, cooling_time_seconds)
simulated_pu241 = result.get(nucname.id("Pu241"), 0.0)
simulated_am241 = result.get(nucname.id("Am241"), 0.0)

# ---
# Calculate the ratios for comparison
# ---
empirical_ratio = pu241_measured / am241_measured
simulated_ratio = simulated_pu241 / simulated_am241

# ---
# Plotting
# ---
plt.figure(figsize=(8, 6))

# Plot empirical and simulated data as single points
plt.scatter(cooling_time_years, empirical_ratio, color='red', marker='x', s=60, linewidths=2,
            label=f"Empirical Ratio (SFCOMPO 2.0) = {empirical_ratio:.2f}")

plt.scatter(cooling_time_years, simulated_ratio, color='blue', marker='o', s=60, linewidths=2,
            label=f"Simulated Ratio (PyNE) = {simulated_ratio:.2f}")

plt.xlabel("Cooling Time (Years)")
plt.ylabel("Pu-241 to Am-241 Isotopic Ratio")
plt.title("Comparison of Empirical vs. Simulated Ratios for MAGNOX Sample A'")
plt.grid(True, which='both', linestyle='--')
plt.legend(loc='best')
plt.tight_layout()
plt.show()

```

```

from pyne.material import Material
from pyne import nucname
import numpy as np

# Define enriched U fuel
# Can be changed to specifics.
# Iteration 1.
fuel = Material({nucname.id('U235'): 0.05, nucname.id('U238'): 0.95})
# Normalized to 1.
fuel.mass = 1.0

# Irradiate the fuel
# Can change flux value.
flux = 1e14
irradiation_time = 365.25 * 24 * 3600
irradiated_fuel = fuel.irradiate(flux=flux, time=irradiation_time)

# Decay the irradiated fuel
cooling_time = 10 * 365.25 * 24 * 3600
cooled_fuel = irradiated_fuel.decay(cooling_time)

# Extract Pu-241 and Am-241 masses
pu241_id = nucname.id("Pu241")
am241_id = nucname.id("Am241")

pu241_mass = cooled_fuel.mass.get(pu241_id, 0.0)
am241_mass = cooled_fuel.mass.get(am241_id, 0.0)

# Output the results
print(f"Pu-241 mass: {pu241_mass:.6e} g")
print(f"Am-241 mass: {am241_mass:.6e} g")

if am241_mass > 0:
    print(f"Pu-241 / Am-241 ratio: {pu241_mass / am241_mass:.4f}")
else:
    print("Am-241 not produced in a measurable quantity.")

```

6.3.Appendix C: Additional Reactor Data

Reactor Type	Sample/Bundle Identifier	Discharge Date	Measurement Date	Cooling Time (Years)	Burnup (GW·d/tU)	Pu Uncertainty	Am Uncertainty	Pu Isotopic Concentration (g/gU)	Am Isotopic Concentration (g/gU)
RBMK-1000	R-1029 Sample 1	12/8/1978	1/1/1987	8.07	17.7	0.98%	0.98%	5.11E-04	2.50E-08
RBMK-1000	REV-34 Sample 5	5/1/1981	1/1/1987	5.67	16.52	0.88%	0.88%	4.55E-04	2.00E-08
RBMK-1000	R-870 (Sample 9)	10/7/1976	1/1/1987	10.23	12.37	1.59%	1.59%	3.15E-04	1.00E-08
RBMK-1000	RG-221 Sample 10	7/14/1977	1/1/1987	9.47	6.81	1.71%	1.71%	1.17E-04	3.00E-09
RBMK-1000	REA-35-74 (Sample 18)	10/1/1980	1/1/1987	6.25	23.8	0.65%	0.65%	6.17E-04	3.00E-08
RBMK-1000	REB-01-74 (Sample 19)	2/8/1981	1/1/1987	5.89	22.4	0.70%	0.70%	5.70E-04	2.50E-08
RBMK-1000	R1-20-2603 Sample 20	5/1/1979	1/1/1987	7.67	6.2	1.33%	1.33%	7.50E-05	1.00E-09
RBMK-1000	R1-20-7788 Sample 28	9/30/1981	1/1/1987	5.25	10.33	0.46%	0.46%	2.16E-04	1.50E-08
RBMK-1000	4-20-3675-85 Sample 35	2/28/1987	1/1/1991	3.84	9.91	0.99%	0.99%	2.02E-04	1.50E-08
RBMK-1000	4-20-3675-85 Sample 40	12/1/1989	1/1/1992	2.08	12.98	0.93%	0.93%	3.23E-04	1.00E-08
MAGNOX (Hunterston)	Sample A'	8/27/1976	3/9/1977	0.53	3.271	1.33%	2.26%	7.12E-05	8.85E-06
MAGNOX (Hunterston)	Sample B'	8/20/1976	3/9/1977	0.55	6.576	2.49%	2.34%	3.15E+04	2.10E-04
CANDU	PK-1 PA19558C 7 1	-	-	15.3	0.221	20.00%	9.00%	1.34E-4	1.466E-4

